AIMS AND GOALS AT HACKERFAB IITB.

I am a first-year Electrical Engineering student, and what draws me to HackerFab is a simple fact that I still find fascinating: we have built the entire digital world and reached our computing capabilities by taking silicon, one of the most common materials on Earth, and learning to control its properties to make switches that can represent and process information. I want to understand this from the ground up, by making such devices myself and to eventually make simple (but cool) circuits that can do something basic such as automatic night light using photodiode and logic gates, or a touch activated switch using a MOS capacitor and logic gates etc.

I also believe that coursework alone cannot give that understanding. We truly learn a field when we build something, watch it fail, debug it, and then push it further. HackerFab's philosophy of cheap, open tools that allows students like me to break and improve upon such machines matches how I want to learn.

My short-term goals at HackerFab IITB are to understand and try out the process flow hands-on and in detail namely cleaning the wafer, thermal oxidation of the wafer in the tube furnace, applying photoresist, lithography, doping, sputtering etc. Along with this I aim to understand the physics of how all these semiconductor devices work and learn to simulate them on tools like LTspice and NGspice to compare the actual lab readings with the ideal case.

My long-term goals include making NMOS MOSFET reliably starting with MOS capacitors and moving on to simple logic gates using these like the NOT, NAND and the NOR gates, since these are the building blocks of all digital circuits. Another area I wish to work upon is to try and scale down the lithography process from tens of micrometres to just a couple of micrometres by understanding the problems and the current limitations of the lithography process.

My main aim at HackerFab is to learn new things and enjoy my time there!!!